

Unit Testing in Java Using the JUnit Framework

The JUnit Framework can be easily integrated with Eclipse. Unit testing accelerates programming speed.



While software testing is generally performed by the professional software tester, unit testing is often performed by the software developers working on the project, at that point of time. Unit testing ensures that the specific function is working perfectly. It also reduces software development risks, cost and time. Unit testing is performed by the software developer during the construction phase of the software development life cycle. The major benefit of unit testing is that it reduces the construction errors during software development, thus improving the quality of the product. Unit testing is about testing classes and methods.

What is JUnit?

JUnit is a testing tool for the Java programming language. It is very helpful when you want to test each unit of the project during the software development process.

How to perform unit testing using the JUnit testing tool

To perform JUnit testing in Java, first of all, you have to install the Eclipse editor. Installation of the latest version is recommended. You can download the Eclipse IDE from the following link: <http://eclipse.org/downloads/>.

In the Eclipse editor, you can write any code. For example, let's suppose that I want to test the following code:

Code-1

```
1 package com;
2
3 public class Junit {
```

```
4
5     public String concatenate(String firstName, String
        lastName) {
6
7         return firstName + lastName;
8     }
9
10    public int multiply(int number1, int number2) {
11
12        return number1 * number2;
13    }
14
15 }
```

After writing Code-1, let's write two test cases—one for the *concatenate* method and the other for the *multiply* method of the JUnit class defined in this code. To create JUnit test cases, you need to click on the Eclipse editor: *File→New→JUnit Test Case*

Defining the test case for the *concatenate()* method of the JUnit class (Code-1)

Code-2

```
1 package com;
2 import static org.junit.Assert.*;
3 import org.junit.Test;
4
5 public class ConcatTest {
6
```